

```
!pip install pymongo
```

```
Requirement already satisfied: pymongo in /usr/local/lib/python3.12/dist-  
Requirement already satisfied: dnspython<3.0.0,>=2.6.1 in /usr/local/lib/
```

```
import os  
for file in os.listdir('/content'):  
    if file.endswith('.csv'):  
        print(file)
```

24-hours Delhi Power Consumption dataset.csv

```
import pandas as pd
```

```
df = pd.read_csv('/content/24-hours Delhi Power Consumption dataset.csv')  
df.head()
```

	Unnamed: 0	timestamp	day_of_week	hour_of_day	is_weekend	temperature
0	0	2023-01-01 00:00:00	6	0	1	17.483571
1	1	2023-01-01 01:00:00	6	1	1	14.308678
2	2	2023-01-01 02:00:00	6	2	1	18.238443
3	3	2023-01-01 03:00:00	6	3	1	22.615149
4	4	2023-01-01 04:00:00	6	4	1	13.829233

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
df.shape  
df.columns.tolist()  
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 8737 entries, 0 to 8736  
Data columns (total 10 columns):  
#   Column                Non-Null Count  Dtype  
---  -  
0   Unnamed: 0            8737 non-null   int64  
1   timestamp             8737 non-null   object
```

```
2    day_of_week      8737 non-null    int64
3    hour_of_day      8737 non-null    int64
4    is_weekend        8737 non-null    int64
5    temperature      8737 non-null    float64
6    is_holiday        8737 non-null    int64
7    solar_generation  8737 non-null    float64
8    load              8737 non-null    int64
9    compensation_method 8737 non-null    object
dtypes: float64(2), int64(6), object(2)
memory usage: 682.7+ KB
```

```
!pip install pymongo
```

```
Requirement already satisfied: pymongo in /usr/local/lib/python3.12/dist-
Requirement already satisfied: dnspython<3.0.0,>=2.6.1 in /usr/local/lib/
```

```
!pip install certifi
```

```
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-
```

```
from pymongo import MongoClient
import certifi

connection_string = "mongodb+srv://navya1231234_db_user:navya2057@electr
client = MongoClient(connection_string, tlsCAFile=certifi.where())
result = client.admin.command('ping')
print(result)
```

```
{'ok': 1}
```

```
db = client['electricity_db']
collection = db['raw_consumption']

data = df.to_dict('records')
collection.insert_many(data)

print("Total documents inserted:", collection.count_documents({}))
```

```
Total documents inserted: 8737
```

```
# MongoDB se raw data wapas nikaalo
raw_data = pd.DataFrame(list(collection.find()))

# MongoDB ka extra '_id' column hata do
raw_data = raw_data.drop(columns=['_id'])

raw_data.head()
raw_data.shape
```

(8737, 10)

```
raw_data['timestamp'] = pd.to_datetime(raw_data['timestamp'])
raw_data['year'] = raw_data['timestamp'].dt.year
raw_data['month'] = raw_data['timestamp'].dt.month
raw_data.head()
raw_data.dtypes
```

	0
Unnamed: 0	int64
timestamp	datetime64[ns]
day_of_week	int64
hour_of_day	int64
is_weekend	int64
temperature	float64
is_holiday	int64
solar_generation	float64
load	int64
compensation_method	object
year	int32
month	int32

dtype: object

```
print("Duplicates:", raw_data.duplicated().sum())
print("Missing values:\n", raw_data.isnull().sum())
```

```
Duplicates: 0
Missing values:
  Unnamed: 0      0
timestamp      0
day_of_week     0
hour_of_day     0
is_weekend      0
temperature     0
is_holiday      0
solar_generation 0
load            0
compensation_method 0
year            0
month           0
dtype: int64
```

```
!pip install sqlalchemy psycopg2-binary
```

Requirement already satisfied: sqlalchemy in /usr/local/lib/python3.12/dist-packages
Collecting psycopg2-binary

Downloading psycopg2_binary-2.9.12-cp312-cp312-manylinux2014_x86_64.manylinux2014_x86_64.whl (4.3 MB)

Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.12/dist-packages (from psycopg2-binary)

Requirement already satisfied: typing-extensions>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from greenlet>=1->psycopg2-binary)

Downloading psycopg2_binary-2.9.12-cp312-cp312-manylinux2014_x86_64.manylinux2014_x86_64.whl (4.3 MB)

4.3/4.3 MB 36.9 MB/s

Installing collected packages: psycopg2-binary

Successfully installed psycopg2-binary-2.9.12

```
!pip install sqlalchemy psycopg2-binary
```

Requirement already satisfied: sqlalchemy in /usr/local/lib/python3.12/dist-packages

Requirement already satisfied: psycopg2-binary in /usr/local/lib/python3.12/dist-packages (from sqlalchemy)

Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.12/dist-packages (from psycopg2-binary->sqlalchemy)

Requirement already satisfied: typing-extensions>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from greenlet>=1->psycopg2-binary->sqlalchemy)

```
from sqlalchemy import create_engine
```

```
connection_string = "postgresql://postgres:soqrnqefhfhthdoajwnzz:k7u%2F6g@localhost:5432/postgres"
```

```
engine = create_engine(connection_string)
```

```
raw_data.to_sql('consumption_data', engine, if_exists='replace', index=False)  
print("Data successfully loaded to Supabase!")
```

Data successfully loaded to Supabase!

```
from scipy import stats
```

```
weekday = raw_data[raw_data['is_weekend']==0]['load']
```

```
weekend = raw_data[raw_data['is_weekend']==1]['load']
```

```
t_stat, p_value = stats.ttest_ind(weekday, weekend)
```

```
print("T-statistic:", t_stat)
```

```
print("P-value:", p_value)
```

T-statistic: -7.438504580522212

P-value: 1.1146450671395673e-13

```
!pip install gspread gspread_dataframe
```

Requirement already satisfied: gspread in /usr/local/lib/python3.12/dist-packages

Requirement already satisfied: gspread_dataframe in /usr/local/lib/python3.12/dist-packages (from gspread)

Requirement already satisfied: google-auth>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from gspread)

Requirement already satisfied: google-auth-oauthlib>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from google-auth>=1.12.0->gspread)

Requirement already satisfied: pandas>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from gspread_dataframe)

Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from google-auth>=1.12.0->gspread)

Requirement already satisfied: pyasn1-modules>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from google-auth>=1.12.0->gspread)

```
Requirement already satisfied: cryptography>=38.0.3 in /usr/local/lib/pyt
Requirement already satisfied: rsa<5,>=3.1.4 in /usr/local/lib/python3.12
Requirement already satisfied: requests-oauthlib>=0.7.0 in /usr/local/lib
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/p
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.1
Requirement already satisfied: cffi>=2.0.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in /usr/local/lib/pyt
Requirement already satisfied: oauthlib>=3.0.0 in /usr/local/lib/python3.
Requirement already satisfied: requests>=2.0.0 in /usr/local/lib/python3.
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dis
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/pytho
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/pytho
```

```
from google.colab import auth
auth.authenticate_user()

import gspread
from google.auth import default

creds, _ = default()
gc = gspread.authorize(creds)

print("Authenticated successfully!")
```

Authenticated successfully!

```
sh = gc.create('Electricity_Consumption_Data')
worksheet = sh.get_worksheet(0)

from gspread_dataframe import set_with_dataframe
set_with_dataframe(worksheet, raw_data)

print("Data uploaded to Google Sheets!")
print("Sheet URL:", sh.url)
```

Data uploaded to Google Sheets!
Sheet URL: <https://docs.google.com/spreadsheets/d/1X16D7mTMx9DNBjUKhL092g>

```
## Step 1: Data Loading & Cleaning
## Step 2: MongoDB (NoSQL) - Raw Data Storage
## Step 3: ETL - Transform & Clean
## Step 4: SQL (Supabase) - Structured Storage
## Step 5: Statistical Analysis (Hypothesis Testing)
## Key Insights
```

·driven demand rather than AC-driven cooling load